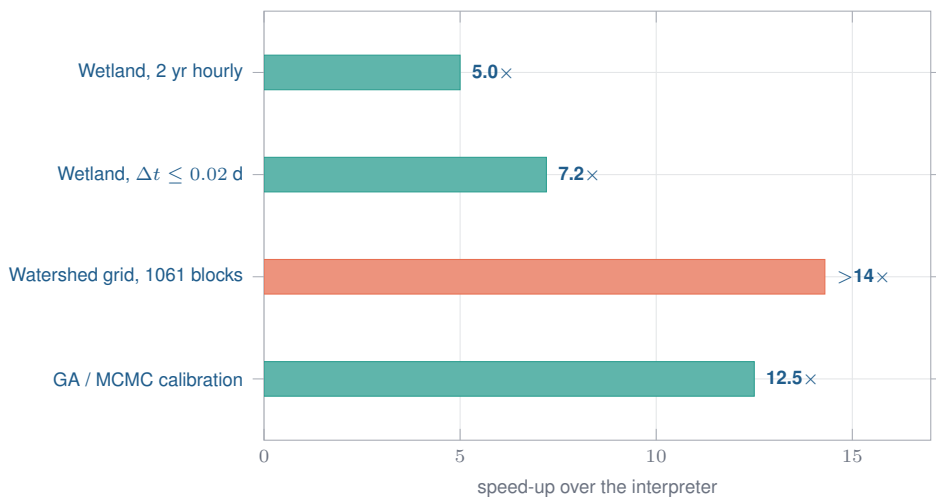


builds on a machine that has never
seen OpenHydroQual



Absolute times. Wetland, two years of hourly forcing: **31 s** interpreted, **6.2 s** compiled. Watershed grid: the dense interpreted solve **timed out past 15 min**, the compiled model finished in **62.9 s** single-threaded. Calibration is typically 10–15×, which is where it matters most: thousands of repeated solves.

Agreement. 0.13% RMS on storages and 0.2% RMS on tracer mass (wetland); 991 of 1061 blocks within 10^{-6} and 0.75% total mass difference (grid).